

AI Data Ingestion & Security Policy

Sentinel-RAG Pipeline | Classification: INTERNAL

0. Definitions

Knowledge Base:

Static threat intelligence documents embedded in the vector database, including CVE records, CWE weakness definitions, and MITRE ATT&CK mappings.

Operational Telemetry:

Raw, unlabeled server logs generated in JSON or CSV format, representing application and network activity.

Ground Truth:

A separate validation artifact mapping event identifiers to attack types. Ground truth is never exposed to the LLM.

LLM:

A Large Language Model used exclusively for contextual threat analysis and explanation within a Retrieval-Augmented Generation (RAG) pipeline.

1. Purpose

This policy defines the authorized data sources, schemas, and security boundaries for data ingestion into the Sentinel-RAG pipeline. The objective is to ensure that the LLM operates solely on approved threat intelligence and controlled telemetry, preventing data leakage, label contamination, or unauthorized system access.

2. Scope

This policy applies to the Data Ingestion and Retrieval layers of the Sentinel-RAG architecture.

Covered Components:

- Vector Database (Threat Intelligence Knowledge Base)
- Telemetry Preprocessing Pipeline
- LLM Inference Context Assembly

Supported Formats:

- JSON
- CSV

Update Frequency:

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- Knowledge Base: Static, version-controlled
- Telemetry: Dynamic, per inference request

3. Authorized Material for Ingestion

3.1 Category A: Threat Intelligence Knowledge Base (Embedded Context)

The vector database may embed and retrieve only curated, non-operational threat intelligence documents, including:

- Vulnerability Definitions (CVE): SQL Injection, XSS, Path Traversal, Command Injection.
- Authentication & Access Attacks: Brute Force, Unauthorized Access.
- Availability Attacks: Denial of Service (DoS/DDoS).
- Reconnaissance Techniques: Network and Port Scanning.
- Weakness Enumeration (CWE): CWE-89, CWE-79, CWE-22, CWE-78, CWE-307, CWE-400, CWE-863, CWE-200.
- MITRE ATT&CK Mappings: T1190 (Exploit Public-Facing Application), T1059 (Command and Scripting Interpreter), T1110 (Brute Force), T1499 (Denial of Service), T1078 (Valid Accounts), T1046 (Network Service Discovery).

3.2 Category B: Operational Telemetry (Inference-Time Context Only)

The LLM may process raw telemetry records transiently during inference. Authorized telemetry fields include:

- Event Metadata: event_id (UUID), timestamp (ISO 8601).
- Network Context: source_ip, dest_port, protocol, connection_result.
- HTTP Context: http_method, endpoint, response_code, user_agent.
- Authentication Context: username (synthetic), auth_result, failure_reason.
- Payload Indicators: query_params (potential attack vectors).
- Performance Metrics: response_time_ms.

3.3 RAG and LLM Security Boundaries

Raw telemetry is never embedded, indexed, or persisted in the vector database.

Only approved threat intelligence documents are embedded for retrieval.

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Operational telemetry is supplied ephemerally to the LLM at inference time and is discarded immediately after response generation.

The LLM has no direct access to production systems, live databases, external networks, or data sources outside this policy.

3.4 Data Provenance and Synthetic Data Declaration

All operational telemetry ingested into the Sentinel-RAG pipeline is synthetically generated for research and training purposes. The dataset contains no real user data, no production traffic, and no logs originating from live systems.

4. Prohibited Material (Exclusion List)

The following materials are strictly prohibited from ingestion into the Knowledge Base or LLM context:

- Ground Truth Labels or event-to-attack mappings.
- Production secrets, API keys, passwords, tokens, or cryptographic material.
- Personally Identifiable Information (PII) or real user credentials.
- Decrypted network traffic or full request/response bodies from live systems.

5. Change Control

Policy Version: 1.1

Release Date: 2025-12-18

Change Log:

- v1.0: Initial release
- v1.1: Alignment with synthetic telemetry schema and expanded attack coverage

All future changes must document date, author, and summary of modifications.

6. Contact Information

For questions, audits, or clarification regarding this policy, contact:

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Security Architecture Team

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